

COMPTE RENDU

Projet Réseau & Sécurité

Infrastructure réseau complète avec VLANs, FortiGate, DMZ et serveur web

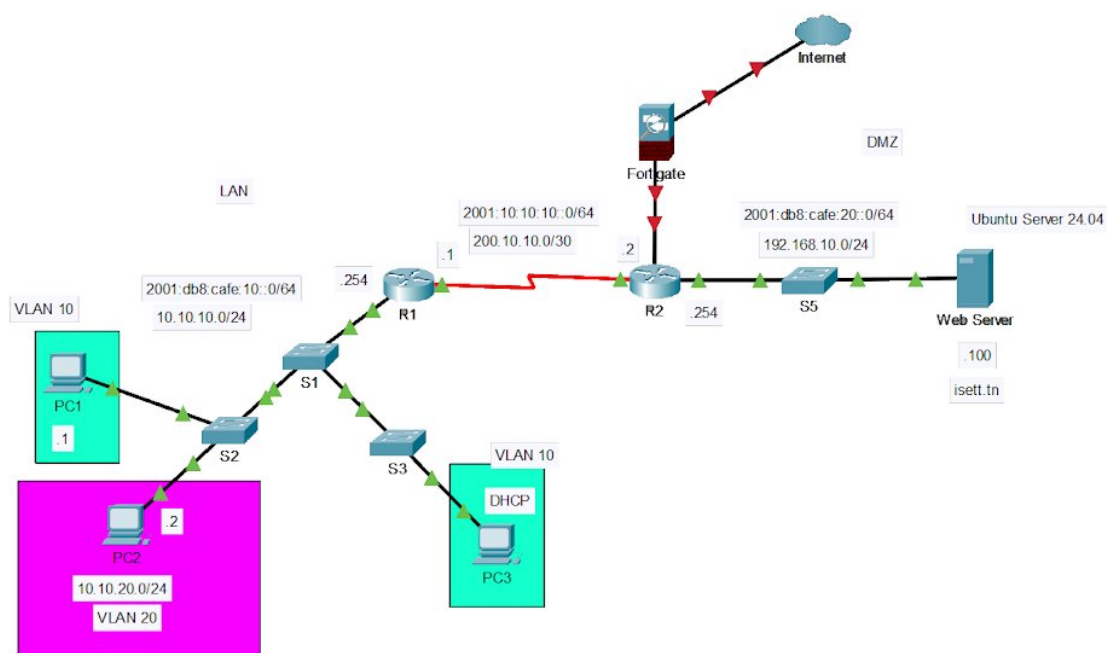


Figure 1 — Topologie réseau du projet (GNS3)

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Encadreur	Mr. Wahid Hamdi
Etablissement	ISETT Tozeur
Filiere	Licence RSI — Réseaux & Systemes Informatiques

Date	19 Mai 2026
Simulation	GNS3 avec FortiGate 7.0.9
Site web DMZ	http://isett.tn

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1. Introduction & Objectifs

Ce compte rendu presente la mise en oeuvre d'une infrastructure reseau complete simulee sous GNS3. Le projet couvre la segmentation par VLANs, le routage inter-VLAN, la securisation par pare-feu FortiGate, la creation d'une zone demilitarisee (DMZ) hebergeant un serveur web Apache2 Ubuntu 24.04, ainsi que la connectivite IPv4 et IPv6.

Objectif	Detail
VLANs (10 & 20)	Segmentation du LAN en deux sous-reseaux isoles
Routage inter-VLAN	Router-on-a-stick sur R1 (subinterfaces Fa0/0.10 / .20)
FortiGate	Pare-feu avec NAT, DMZ, politiques de filtrage
DMZ	Serveur Apache2 accessible depuis le WAN (HTTP/HTTPS)
IPv6	Adressage dual-stack sur R1 (2001:db8:cafe::/64)

2. Topologie du reseau

La topologie ci-dessous montre l'ensemble des équipements interconnectés dans GNS3 : deux routeurs Cisco, cinq switches, un pare-feu FortiGate, trois PCs (VPCS) et un serveur Ubuntu.

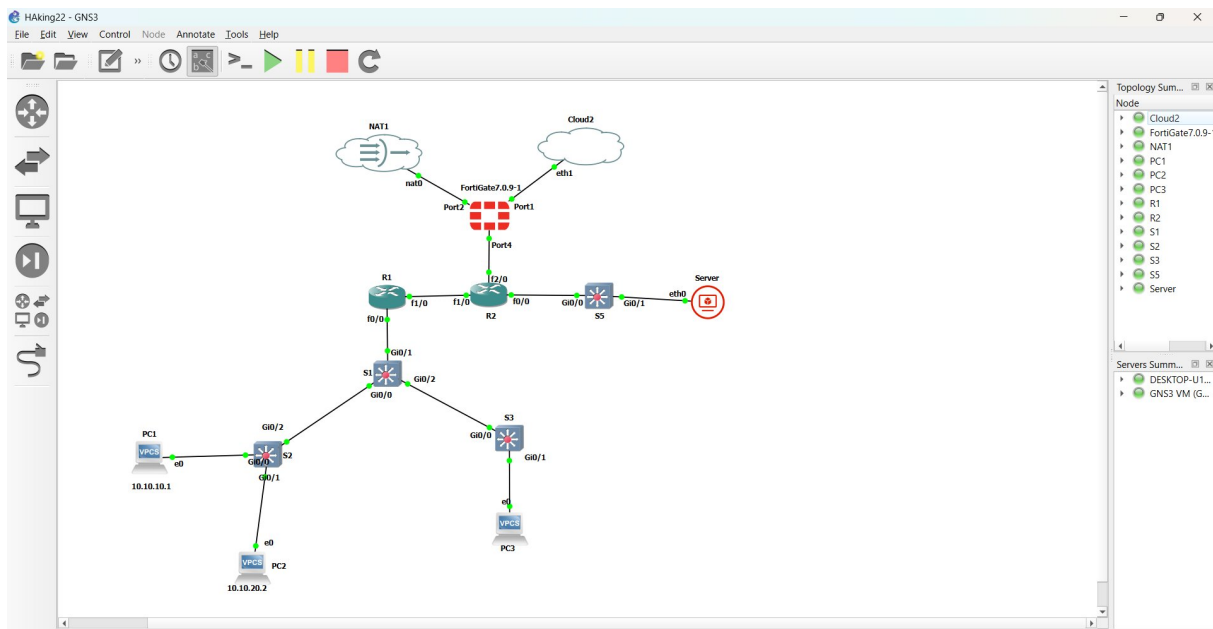


Figure 2 — Capture GNS3 : topologie complete du projet

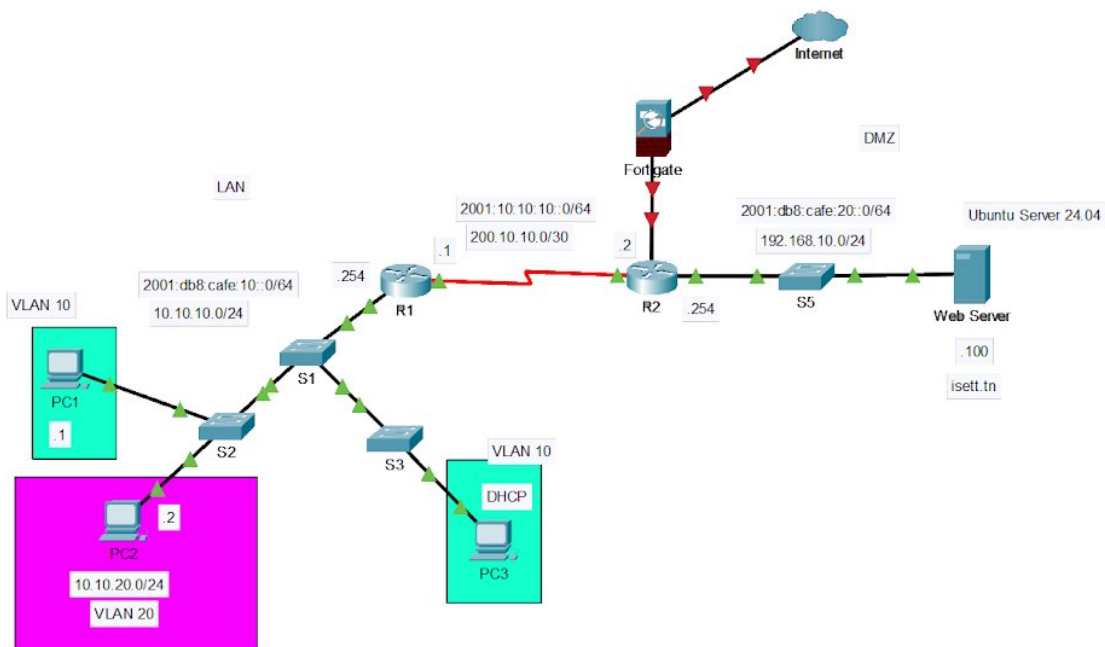


Figure 3 — Schema logique avec adressage IP et VLANs

3. Plan d'adressage IP

Equipement	Interface	Adresse IPv4	Adresse IPv6	Role
R1	Fa0/0.10	10.10.10.254/24	2001:db8:cafe:10::1/64	Gateway VLAN10
R1	Fa0/0.20	10.10.20.254/24	2001:db8:cafe:20::254/64	Gateway VLAN20
R1	Fa1/0	10.0.0.1/30	—	WAN vers FortiGate
FortiGate	port3	10.0.0.2/30	—	LAN (vers R1)
FortiGate	port4	172.16.0.1/30	—	DMZ (vers R2)
FortiGate	port2	192.168.48.2/24	—	Management
FortiGate	port1	192.168.230.139	—	WAN / Internet
R2	Fa0/0	192.168.10.254/24	—	Gateway DMZ
R2	Fa1/0	10.0.0.2/30	—	WAN (vers FortiGate)
R2	Fa2/0	172.16.0.2/30	—	Lien FortiGate DMZ
Web Server	ens37	192.168.48.135/24	—	Serveur Apache2
PC1	e0	10.10.10.1/24	—	VLAN 10
PC2	e0	10.10.20.2/24	—	VLAN 20
PC3	e0	DHCP	—	VLAN 10

4. Configuration R1 — Routeur LAN

R1 assure le routage inter-VLAN via des sous-interfaces (router-on-a-stick) et gere le dual-stack IPv4/IPv6 pour le LAN.

4.1 show ip interface brief + show ip route

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          unassigned      YES NVRAM  up          up
FastEthernet0/0.10       10.10.10.254    YES NVRAM  up          up
FastEthernet0/0.20       10.10.20.254    YES NVRAM  up          up
FastEthernet1/0          10.0.0.1        YES NVRAM  up          up
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.0.2
      10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C      10.0.0.0/30 is directly connected, FastEthernet1/0
L      10.0.0.1/32 is directly connected, FastEthernet1/0
C      10.10.10.0/24 is directly connected, FastEthernet0/0.10
L      10.10.10.254/32 is directly connected, FastEthernet0/0.10
C      10.10.20.0/24 is directly connected, FastEthernet0/0.20
L      10.10.20.254/32 is directly connected, FastEthernet0/0.20
      172.16.0.0/30 is subnetted, 1 subnets
S      172.16.0.0 [1/0] via 10.0.0.2
R1#
```

Figure 4 — R1 : interfaces IP et table de routage IPv4

4.2 show ipv6 interface brief

```

R1#show ipv6 interface brief
FastEthernet0/0          [up/up]
    unassigned
FastEthernet0/0.10       [up/up]
    FE80::C801:1FFF:FE8C:0
FastEthernet0/0.20       [up/up]
    FE80::C801:1FFF:FE8C:0
    2001:DB8:CAFE:20::254
FastEthernet1/0          [up/up]
    FE80::C801:1FFF:FE8C:1C
    2001:DB8:CAFE:10::1
    2001:DB8:CAFE:10::2
R1#

```

Figure 5 — R1 : interfaces IPv6 (2001:db8:cafe:10::/64 & :20::/64)

4.3 Analyse de la table de routage

Route	Type	Description
0.0.0.0/0 via 10.0.0.2	S*	Route par défaut vers FortiGate
10.10.10.0/24 — Fa0/0.10	C	VLAN 10 directement connecte
10.10.20.0/24 — Fa0/0.20	C	VLAN 20 directement connecte
10.0.0.0/30 — Fa1/0	C	Lien WAN vers FortiGate
172.16.0.0/30 via 10.0.0.2	S	Reseau DMZ (FortiGate port4)

5. Configuration R2 — Routeur DMZ

R2 est le routeur de la zone DMZ. Il achemine le trafic entre FortiGate et le serveur web (192.168.10.0/24) et assure la route par défaut vers FortiGate.

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 172.16.0.1 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 172.16.0.1
     10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C     10.0.0.0/30 is directly connected, FastEthernet1/0
L     10.0.0.2/32 is directly connected, FastEthernet1/0
S     10.10.10.0/24 [1/0] via 10.0.0.1
S     10.10.20.0/24 [1/0] via 10.0.0.1
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.16.0.0/30 is directly connected, FastEthernet2/0
L     172.16.0.2/32 is directly connected, FastEthernet2/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.10.0/24 is directly connected, FastEthernet0/0
L     192.168.10.254/32 is directly connected, FastEthernet0/0
R2#
R2#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.10.254  YES NVRAM   up          up
FastEthernet1/0          10.0.0.2        YES NVRAM   up          up
FastEthernet2/0          172.16.0.2      YES NVRAM   up          up
R2#
```

Figure 6 — R2 : table de routage et interfaces (DMZ)

6. FortiGate — Interfaces reseau

Le FortiGate 7.0.9 dispose de quatre ports actifs avec des roles bien definis : WAN (port1), Management (port2), LAN/R1 (port3), DMZ/R2 (port4).

```
FortiGate-VM64-KVM # get system interface
== [ port1 ]
name: port1  mode: dhcp  ip: 192.168.230.139 255.255.255.0  status: up  ne
tbios-forward: disable  type: physical  ring-rx: 0  ring-tx: 0  netflow-sam
pler: disable  sflow-sampler: disable  src-check: enable  explicit-web-pro
xy: disable  explicit-ftp-proxy: disable  proxy-captive-portal: disable  m
tu-override: disable  wccp: disable  drop-overlapped-fragment: disable  dr
op-fragment: disable
== [ port2 ]
name: port2  mode: static  ip: 192.168.48.2 255.255.255.0  status: up  net
bios-forward: disable  type: physical  ring-rx: 0  ring-tx: 0  netflow-sam
pler: disable  sflow-sampler: disable  src-check: enable  explicit-web-prox
y: disable  explicit-ftp-proxy: disable  proxy-captive-portal: disable  mt
u-override: disable  wccp: disable  drop-overlapped-fragment: disable  dro
p-fragment: disable
== [ port3 ]
name: port3  mode: static  ip: 10.0.0.2 255.255.255.252  status: up  netbi
os-forward: disable  type: physical  ring-rx: 0  ring-tx: 0  netflow-sample
r: disable  sflow-sampler: disable  src-check: enable  explicit-web-proxy:
disable  explicit-ftp-proxy: disable  proxy-captive-portal: disable  mtu-
override: disable  wccp: disable  drop-overlapped-fragment: disable  drop-
fragment: disable
== [ port4 ]
name: port4  mode: static  ip: 172.16.0.1 255.255.255.252  status: up  net
bios-forward: disable  type: physical  ring-rx: 0  ring-tx: 0  netflow-sam
pler: disable  sflow-sampler: disable  src-check: enable  explicit-web-prox
y: disable  explicit-ftp-proxy: disable  proxy-captive-portal: disable  mt
u-override: disable  wccp: disable  drop-overlapped-fragment: disable  dro
p-fragment: disable
```

Figure 7 — FortiGate : get system interface

Port	Mode	Adresse IP	Role
port1	DHCP	192.168.230.139/24	WAN — acces Internet
port2	Static	192.168.48.2/24	Management
port3	Static	10.0.0.2/30	LAN — connexion vers R1
port4	Static	172.16.0.1/30	DMZ — connexion vers R2

7. FortiGate — Table de routage

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

Routing table for VRF=0
S*    0.0.0.0/0 [5/0] via 192.168.230.2, port1, [1/0]
S     10.0.0.0/30 [10/0] via 172.16.0.2, port4, [1/0]
S     10.10.10.0/24 [10/0] via 172.16.0.2, port4, [1/0]
S     10.10.20.0/24 [10/0] via 172.16.0.2, port4, [1/0]
C     172.16.0.0/30 is directly connected, port4
S     192.168.10.0/24 [10/0] via 172.16.0.2, port4, [1/0]
C     192.168.48.0/24 is directly connected, port2
C     192.168.230.0/24 is directly connected, port1
```

Figure 8 — FortiGate : get router info routing-table all

Destination	Type	Via / Interface	Description
0.0.0.0/0	S*	192.168.230.2 — port1	Route par défaut WAN/Internet
10.0.0.0/30	S	172.16.0.2 — port4	Reseau WAN R1-FortiGate
10.10.10.0/24	S	172.16.0.2 — port4	VLAN 10 (LAN)
10.10.20.0/24	S	172.16.0.2 — port4	VLAN 20 (LAN)
172.16.0.0/30	C	port4 (direct)	Lien FortiGate-R2
192.168.10.0/24	S	172.16.0.2 — port4	Reseau DMZ serveur web
192.168.48.0/24	C	port2 (direct)	Management
192.168.230.0/24	C	port1 (direct)	WAN

8. FortiGate — Politiques de securite (Firewall Policy)

Quatre politiques de filtrage ont ete configurees pour controler les flux entre les zones reseau.

```
config firewall policy
  edit 1
    set name "LAN-to-WAN"
    set uuid 4c06de94-509c-51f1-20b4-b4864a4e16d6
    set srcintf "port4"
    set dstintf "port1"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set nat enable
  next
  edit 2
    set name "LAN-to-DMZ"
    set uuid 4cb2eb80-509c-51f1-566a-3424d5dc0f78
    set srcintf "port3"
    set dstintf "port4"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
  next
  edit 3
    set name "DMZ-to-WAN"
--More--
    set uuid 4d42225a-509c-51f1-b0f6-3abcd2b8bd2d
    set srcintf "port4"
    set dstintf "port1"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set nat enable
  next
  edit 4
    set name "WAN-to-DMZ"
    set uuid 4db8397c-509c-51f1-142e-83f4f16828cf
    set srcintf "port1"
    set dstintf "port4"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "HTTP" "HTTPS"
  next
```

Figure 9 — FortiGate : config firewall policy

ID	Nom	Source	Destination	Services	NAT	Action
----	-----	--------	-------------	----------	-----	--------

1	LAN-to-WAN	port4	port1	ALL	Oui	Accept
2	LAN-to-DMZ	port3	port4	ALL	Non	Accept
3	DMZ-to-WAN	port4	port1	ALL	Oui	Accept
4	WAN-to-DMZ	port1	port4	HTTP/HTTPS	Non	Accept

La politique WAN-to-DMZ (edit 4) autorise uniquement HTTP et HTTPS depuis Internet vers le serveur web, ce qui constitue la règle de sécurité principale de la DMZ.

9. Serveur Web Ubuntu 24.04 — Apache2

Le serveur web Ubuntu 24.04 héberge le site isett.tn dans la zone DMZ. Apache2 est configuré et actif en écoute sur le port 80.

```
server@server-VMware-Virtual-Platform:~$ sudo systemctl status apache2
[sudo: authenticate] Password:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
   Active: active (running) since Tue 2026-05-19 20:36:40 CET; 5min ago
   Invocation: e1454e57f18f4e92a403b47d82887805
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1336 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes>
     Tasks: 55 (limit: 3966)
    Memory: 8.4M (peak: 8.6M)
       CPU: 932ms
    CGroup: /system.slice/apache2.service
            └─1336 /usr/sbin/apache2 -k start -DFOREGROUND
              └─1422 /usr/sbin/apache2 -k start -DFOREGROUND
                └─1423 /usr/sbin/apache2 -k start -DFOREGROUND

May 19 20:36:33 server-VMware-Virtual-Platform systemd[1]: Starting apache2.serv>
May 19 20:36:40 server-VMware-Virtual-Platform apachectl[1336]: AH00558: apache>
May 19 20:36:40 server-VMware-Virtual-Platform systemd[1]: Started apache2.serv>
lines 1-18/18 (END)
```

Figure 10 — Serveur : systemctl status apache2 (active/running)

```
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:a3:19:ce brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29a319ce
    inet 192.168.230.10/24 brd 192.168.230.255 scope global noprefixroute ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fea3:19ce/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:a3:19:d8 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c29a319d8
    inet 192.168.48.135/24 brd 192.168.48.255 scope global dynamic noprefixroute
ens37
    valid_lft 1598sec preferred_lft 1598sec
    inet6 fe80::f2e1:829a:a52e:7d1c/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
server@server-VMware-Virtual-Platform:~$
```

Figure 11 — Serveur : ip addr show (192.168.48.135/24 & 192.168.230.10/24)

Le service apache2 est actif depuis le 19 Mai 2026 a 20h36 CET. PID principal : 1336. Memoire : 8.4 Mo.
Trois workers sont en cours d'execution.

10. Site web isett.tn — CV en ligne

Le site web héberge un CV professionnel interactif de Khalifa Yassin, accessible via <http://isett.tn> depuis le réseau interne.

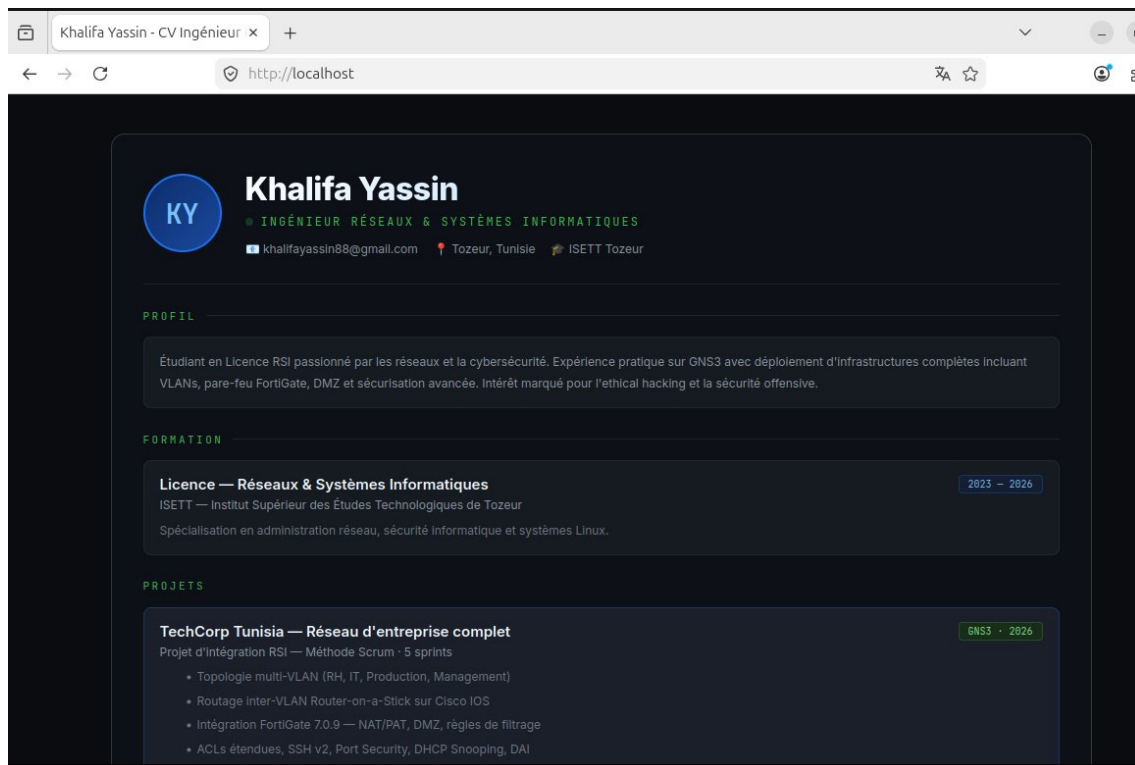


Figure 12 — Site web <http://isett.tn> : CV Khalifa Yassin (Ingenieur Reseaux & Systemes)

Le CV présente une formation en Licence RSI à l'ISETT Tozeur (2023-2026) avec spécialisation en administration réseau, sécurité et systèmes Linux. Le projet TechCorp Tunisia y est documenté : topologie multi-VLAN, FortiGate, DMZ, ACLs et DHCP Snooping.

11. Conclusion

Ce projet a permis de mettre en oeuvre une infrastructure reseau complete et fonctionnelle sous GNS3, integrant les technologies cles des reseaux d'entreprise modernes :

Composant	Statut	Remarques
VLANs 10 & 20	OK	Segmentation LAN operationnelle
Routage inter-VLAN (R1)	OK	Router-on-a-stick Fa0/0.10 / .20
IPv6 dual-stack	OK	2001:db8:cafe:10::/64 & :20::/64
FortiGate — NAT	OK	LAN-to-WAN & DMZ-to-WAN avec NAT
FortiGate — DMZ	OK	HTTP/HTTPS autorise depuis WAN
Apache2 (Ubuntu 24.04)	OK	Active/Running, port 80
Site web isett.tn	OK	CV accessible depuis le LAN

L'integration du pare-feu FortiGate avec des politiques de securite granulaires, combinee a la mise en place d'une DMZ pour le serveur web, demontre une approche professionnelle de la securite reseau en profondeur (defense in depth).

Realise par Khalifa Yassin Etudiant Licence RSI ISETT Tozeur	Encadreur Mr. Wahid Hamdi Enseignant ISETT Tozeur	Date 19 Mai 2026 ISETT Tozeur Tunisie
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